

Practitioner Dietary Supplement Reference Guide – 3rd Edition

dotFIT® Product Manufacturing and Testing

InvaPharm

Primary manufacturer of solid-dose dietary supplements, including tablets and capsules.

InvaPharm has a strong industry reputation for producing innovative, value-added supplements with a high rate of on-time delivery. The company offers an impressive selection of over 450 always-in-stock private label products, as well as the ability to manufacture many others on a contract basis. Their 41 years of experience have allowed them to build an innovative Product Development team, a strong Quality and Regulatory Department, an in-house laboratory, and an industry-leading Customer Service Department.

Product Development Team

The development team is led by a Ph.D. formulator and includes a nutritionist, Quality Assurance personnel, and others with significant industrial formulation experience. They offer expertise in highly stable probiotics and protein powder formulation and manufacturing. Among other tools, they have access to an ultra-low-humidity environment. It is this team that can play an advisory role on regulatory, legal, and international export issues.

Quality

Since InvaPharm is regulated by local, state, and federal agencies, including the FTC, the California Department of Health, and the U.S. Food and Drug Administration (FDA), it has established large Quality Assurance and Quality Control departments. Their teams have developed and follow an extensive system of Standard Operating Procedures (SOPs) and Good Manufacturing Practices (GMPs) that ensure regulatory compliance at all levels. A critical part of the Quality team's responsibilities is to control the complex processes of the manufacturing systems, including:

- Development and/or participation of product formulations with strict adherence to dotFIT's evidence-based guidelines
- Specification development
- Materials procurement
- Raw material receipt, testing, and release
- All in-process manufacturing systems
- Sample retention
- Document and records control
- Laboratory oversight
- Physical and analytical testing
- Batch reconciliation
- Finished product review and release
- Internal audits
- Personnel training for compliance with internal SOPs and GMPs
- Compliance with all local, state and federal government regulations
- Compliance with NSF Certified for Sport (NSFCS) unique requirements
 - See below under "About NSF" [for more on NSF & NSFCS](#)

Additional 3rd Party Audits

The InvaPharm quality teams have also been central to the company's compliance with the standards and system controls required by external programs such as the Natural Products Association's GMP certification and NSF GMP certification. These are important third-party organizations that provide external measures of quality assurance and compliance.

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Analytical and Physical Testing

InvaPharm conducts physical and analytical tests throughout the entire manufacturing process and utilizes an in-house laboratory to complete the following tests:

- Physical Testing
 - Tablet hardness
 - Tablet thickness
 - Tablet friability
 - Weight variation
 - Color matching
 - Moisture analysis
 - Tap density
 - Disintegration
- Analytical Testing for Identity and/or Potency
 - Raw material identity through FTNIR spectra matching
 - Finished product analytical testing using
 - HPLC
 - Atomic Absorption
 - Titration
 - NIR
 - UV-Vis
- Microbial Testing
 - Absence of E. Coli
 - Absence of Staphylococcus
 - Absence of Salmonella
 - Total Plate Count

Third-Party Product Testing

All products undergo third-party testing by either Eurofins Scientific or the NSF Certified for Sport Program (NSFCS) (see below for more information on NSFCS banned substance testing). The accredited third-party laboratories confirm the potencies of:

- Botanical extract actives
- Water-soluble vitamins
- Fat-soluble vitamins
- Trace minerals
- Macro minerals
- Enzymes
- Amino acids
- ORAC value
- Antioxidants
- Fatty acids
- Specialty compounds
- Nutritional content (carbs, proteins, fats, cholesterol, sodium, etc.)

And the presence/absence of:

- Heavy metals
- Genetically Modified Organisms (PCR)

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Actus Nutrition

Primary powder formulation manufacturer for dotFIT.

Raw Material to Finish Product Testing

Actus Nutrition maintains a strict quality program on all raw materials and finished products. All incoming raw materials undergo microbiological testing, heavy metals testing, contaminant testing, and analysis for assay if applicable. For protein products, all incoming raw materials are specifically tested to ensure protein content is attained in accordance with the specifications. All finished products are retested for micro and heavy metals, along with assays specific to the product type. This is all done via in-house equipment and varies by ingredient. Actus Nutrition currently has the following equipment to perform these tests: HPLC, FTIR, ICP-MS, GC-MS, and more. In-process testing is also conducted via FTIR and HPLC to ensure that a proper blend has been achieved. FTIR provides a fingerprint of the blend, whereas HPLC provides specific details on an analyte. Certified Certificates of Analysis (C of As) accompany every product batch delivery to dotFIT.

Third Party Quality Controls

Good Manufacturing Practices (cGMPs) are regulatory requirements that provide guidelines for necessary processes, procedures, and documentation, assuring the product produced has the identity, strength, composition, quality, and purity it is represented to possess on the product label. NSF conducts regular plant audits of Actus Nutrition's facility to verify compliance with GMPs. GMPs for the current NSF Dietary Supplements Program are included in NSF American National Standard 173-Dietary Supplements, the only American National Standard for Dietary Supplements, and are consistent with the requirements that the FDA has laid out in 21 CFR § 111.

Third-Party Product Testing

Following the in-house testing and third-party audits listed above, specific dotFIT products produced by Actus Nutrition are sent for additional third-party testing performed by Eurofins Scientific and the NSFCS program, as shown below.

The NSFCS Testing and Certification program (see below “About NSF” for more details)

- APPLICATION
 - Formulation
 - Label
 - Ingredient supplier's information
 - Manufacturing facility's information
- TOXICOLOGY REVIEW
 - Label and formulation review and comparison
 - Ingredient review
 - Determine product testing
- FACILITY INSPECTION
 - Good Manufacturing Practices (GMP) audits of production facilities
 - Observations of in-house laboratories
 - Sourcing and traceability procedures
 - Schedule of ingredient supplier audits based on the number of suppliers
- ANNUAL LABORATORY TESTING/ANALYSIS

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- Microbiological
 - Heavy metals
 - Pesticides/herbicides
 - Label content verification
 - Disintegration
 - Banned substances testing based on the number of lots
- PRODUCT CERTIFICATION/LISTING
 - Monitor formulation/ingredient supplier changes
 - Unannounced follow-up audits
 - Marketplace sampling

Additional information can be found here: [NSF Certified for Sport](#)

NSF Certification for Sport Program (NSFCS)

Although all dotFIT products are formulated and manufactured with the same rigor as described above, a select group of dotFIT products undergoes an additional test and third-party certification for an extra level of assurance. A complete line of dotFIT NSF Certified for Sport (NSFCS) products can be found [here](#). In addition to the dotFIT standard of evidence-based programming, formulas, and third-party testing, the addition of the NSFCS process ensures that collegiate and professional athletes are protected from unwarranted suspensions due to banned substances in supplements. Therefore, not only are the contents tested to match the label, but the program also includes a test for banned substances that can creep into products during the manufacturing process due to unprotected, non-segregated mixing rooms or, worse, deliberate spiking of illegal ingredients. Collegiate, professional, other drug-tested athletes and their team managements require this assurance (NSFCS seal) to protect themselves from tainted products commonly found in commercially available dietary supplements in mass-market outlets.^{1,2,3,4,5}

About NSF

To meet the growing demands of athletes, coaches, and all those concerned about safety and banned substances in sports supplements, NSF International created the NSF Certified for Sport® Program. The program's objective is to certify that participating sports supplement manufacturers have met NSF's stringent independent certification process guidelines, which were developed through a consensus process involving regulatory, sports industry, and consumer groups. This program, which focuses primarily on the sports supplement manufacturing and sourcing process, provides key preventive measures to:

- Protect against adulteration of products
- Verify label claims against product contents
- Identify athletic banned substances in the finished product or ingredients

NSF developed and maintains the only accredited American National Standard for certifying dietary supplements, NSF/ANSI Standard 173. NSF's history of independence led to a partnership with the National Football League (NFL) and the NFL Players Association (NFLPA) to develop and administer the NFL/NFLPA Supplement Certification Program, specifically for professional football but used across all sports.

The NSF Prohibited Substances List includes banned substances, identified by leading sports organizations, such as the World Anti-Doping Agency (WADA), the NFL and Major League Baseball (MLB). The NSF Certified for Sport™ Program certifies products and inspects facilities for a range of substances. [Click here](#) for more on the NSF Certification program.

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Products by Manufacturer and Third-Party Testing

Reliance – NSF Certified Facility

All products are third-party tested by Health Level One and Eurofins Scientific. Additional third-party testing by NSF is listed alongside its respective products.

- Women'sMV™ - Multivitamin & Mineral Formula - **Eurofins Scientific**
- SuperCalcium+™ - **NSF**
- BestPlantProtein Chocolate - **Eurofins Scientific**
- Electrolytes - **Health Level One**
- LeanPak 90 - **Eurofins Scientific**

Lief Labs – NSF Certified Facility

Lief Labs' manufacturing services are conducted within two cGMP-certified, state-of-the-art facilities totaling over 220,000 sq. ft. Lief dedicates 25% of its staff to Quality Assurance & Quality Control. All products are manufactured in accordance with the latest FDA cGMP guidelines and in compliance with 21CFR Part 111.

- GlutamineComplex - **NSF**

Actus Nutrition – NSF Certified Facility

With over 77 years in the business, 11 facilities, and serving 38 countries, dotFIT is proud to have Milk Specialties as one of its partners.

- AminoFormula – Lemonade, Blue Raspberry, and Watermelon – **NSF**
- CreatineMonohydrate - Raspberry Lemonade and Unflavored - **NSF**
- ExtremeCreatineXXXL+ - Raspberry Lemonade – **NSF**
- WheySmooth Chocolate, Vanilla, and Strawberry – **NSF**
- FirstString Chocolate and Vanilla - **NSF**
- Pre/Post Workout Formula & Meal Replacement Chocolate and Vanilla - **Eurofins Scientific**
- All Natural WheySmooth Chocolate and Vanilla - **NSF**
- LeanMeal Vanilla and Chocolate - **Eurofins Scientific**
- BestPlantProtein Vanilla - **Eurofins Scientific (In Progress)**

CapTek – NSF Certified Facility

Quality Control & Assurance

- SuperOmega-3 Fish Oil - **NSF**
- Vitamin D-3 – **NSF**

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InvaPharm – NSF Certified Facility

NSF Certified for Sport and cGMP certified Facility, as well as QAI and California State Organic certified manufacturer that operates a 60,000 square foot facility for production and warehousing.

Certifications

Quality Control & Assurance

- Alln1SuperBlend Orange Burst and Pineapple Swirl- **NSF**
- ActiveMV™ - Multivitamin & Mineral Formula - **NSF**
- Over50MV™ - Multivitamin & Mineral Formula - **Eurofins Scientific**
- Vegan50MV™ - Multivitamin & Mineral Formula - **Eurofins Scientific (DISCONTINUED)**
- KidsMV - Multivitamin & Mineral Formula - **Eurofins Scientific**
- SuperiorAntioxidant™ - **NSF**
- DigestiveEnzymes - **Eurofins Scientific**
- WeightLoss&LiverSupport™ - **Eurofins Scientific**
- UltraProbiotic - **Eurofins Scientific (In Progress)**
- CarbRepel® - **Eurofins Scientific**
- ThermAccel™ - **Eurofins Scientific**
- SleepAid - **Eurofins Scientific**
- EnergyAid (Formerly WorkoutExtreme) - **Eurofins Scientific**

GMP Labs – NSF Certified Facility

NSF GMP for Sport Registered Facility and USDA Organic certified manufacturer operates out of two facilities, totaling over 100,000 square feet.

- JointSkinCollagen+ – **NSF**
- AdvancedBrainHealth - **Eurofins Scientific**

DryBev – NSF Certified Facility

- NO7 Preworkout (Formerly NO7Rage³) Lemonade, Blue Raspberry, Mixed Berry - **Eurofins Scientific**

Pro-Amino International

Manufacturer of meal replacement bars and wafers for dotFIT

Pro-Amino's 75,000-square-foot facility is both GMP and SQF certified, boasting over 30 years of experience in the nutrition industry.

Quality Control

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Noble Foods

Manufacturer of meal replacement bars for dotFIT

Quality Control

Noble Foods produces its bars on a Sollich slab line. All productions are temperature-controlled, tightly monitored for consistency and texture, and every single bar is metal detected and weighed on online check-weighers to ensure that they conform to our clients' specifications.

Built in 2007, Noble Foods operates out of a 30,000-square-foot, state-of-the-art facility. It was built according to their specifications and designed and constructed specifically for the food industry. Noble Foods is presently capable of manufacturing over 50 million bars a year.

References

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- ³ <http://www.forbes.com/sites/alexmorrell/2015/03/12/lawsuits-say-protein-powders-lack-protein-ripping-off-athletes>
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